

## KLAWOCK, AK, USA

WMO: 703894

Lat: 55.580N

Lon: 133.075W

Elev: 12

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 97-19

WBAN: 25367

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 17.5       | 20.9 | 2.9                           | 6.4 | 23.4 | 8.3 | 8.4 | 23.6 | 19.4                  | 31.4 | 17.4 | 32.6 | 6.8                   | 30   | 0.597 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 11.7                   | 70.7            | 59.2 | 67.3 | 58.3 | 64.5 | 57.1 | 61.3                | 67.6 | 59.9 | 64.9 | 58.7 | 62.9 | 7.2                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 58.9                            | 74.2 | 62.1 | 57.5 | 70.6 | 61.1 | 56.6 | 68.2 | 60.3 | 27.2          | 67.5 | 26.3 | 65.0 | 25.4 | 62.8 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 16.4          | 13.4 | 11.7 | 12.8                       | 79.3 | 3.8                | 6.4 | 10.0                                               | 83.9 | 7.8        | 87.6 | 5.7        | 91.2 | 3.0        | 95.8 |    |    |
| (5)               |      |      | 11.2                       | 63.9 | 3.8                | 2.2 | 8.5                                                | 65.5 | 6.3        | 66.8 | 4.1        | 68.0 | 1.4        | 69.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 45.6   | 36.5  | 36.1  | 38.0  | 42.8  | 48.8  | 53.8  | 57.0  | 57.8  | 53.6  | 46.5  | 39.6  | 36.3  |
|                                                                        | DBStd       | 9.46   | 7.18  | 5.86  | 5.63  | 4.17  | 4.51  | 3.73  | 3.05  | 2.90  | 3.41  | 4.36  | 5.91  | 6.47  |
|                                                                        | HDD50       | 2355   | 420   | 390   | 373   | 221   | 76    | 5     | 0     | 0     | 8     | 122   | 314   | 426   |
|                                                                        | HDD65       | 7083   | 884   | 810   | 837   | 667   | 501   | 339   | 249   | 226   | 342   | 575   | 763   | 890   |
|                                                                        | CDD50       | 751    | 1     | 0     | 1     | 4     | 40    | 118   | 216   | 240   | 116   | 13    | 1     | 1     |
|                                                                        | CDD65       | 2      | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 1     | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 61     | 0     | 0     | 0     | 1     | 7     | 33    | 4     | 14    | 2     | 0     | 0     | 0     |
|                                                                        | CDH80       | 11     | 0     | 0     | 0     | 0     | 0     | 10    | 0     | 1     | 0     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 4.4    | 5.5   | 4.6   | 4.9   | 4.4   | 4.2   | 4.0   | 3.8   | 3.7   | 3.8   | 4.3   | 4.9   | 5.1   |
|                                                                        | PrecAvg     | 104.8  | 11.5  | 8.6   | 8.4   | 6.4   | 4.6   | 3.8   | 4.8   | 6.7   | 10.7  | 13.4  | 12.9  | 13.0  |
| (15) Precipitation                                                     | PrecMax     | 142.7  | 21.4  | 17.8  | 15.8  | 9.8   | 11.5  | 7.0   | 8.9   | 14.6  | 19.3  | 20.9  | 23.1  | 28.2  |
|                                                                        | PrecMin     | 75.3   | 3.5   | 3.1   | 3.0   | 0.6   | 0.3   | 1.4   | 0.4   | 1.7   | 3.9   | 5.9   | 5.9   | 5.9   |
|                                                                        | PrecStd     | 17.0   | 3.6   | 3.9   | 3.3   | 2.3   | 2.6   | 1.4   | 2.0   | 3.0   | 3.7   | 3.4   | 4.1   | 5.4   |
|                                                                        | PrecStd     | 17.0   | 3.6   | 3.9   | 3.3   | 2.3   | 2.6   | 1.4   | 2.0   | 3.0   | 3.7   | 3.4   | 4.1   | 5.4   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 53.6  | 52.3  | 59.0  | 66.2  | 73.5  | 78.5  | 72.2  | 75.5  | 69.7  | 59.2  | 54.1  | 54.9  |
|                                                                        |             | MCWB   | 45.0  | 44.7  | 47.5  | 49.6  | 53.6  | 60.3  | 61.7  | 63.0  | 58.4  | 52.6  | 49.1  | 47.2  |
|                                                                        | 2%          | DB     | 48.9  | 48.7  | 51.4  | 57.9  | 64.6  | 67.9  | 68.2  | 70.0  | 64.9  | 56.4  | 51.4  | 49.2  |
|                                                                        |             | MCWB   | 45.9  | 43.4  | 43.8  | 47.5  | 52.6  | 57.5  | 59.9  | 60.5  | 56.4  | 51.6  | 48.7  | 45.6  |
|                                                                        | 5%          | DB     | 46.7  | 46.2  | 48.3  | 53.5  | 60.5  | 63.8  | 65.2  | 66.8  | 62.0  | 54.4  | 49.6  | 46.9  |
|                                                                        |             | MCWB   | 44.4  | 42.2  | 43.0  | 45.7  | 51.5  | 55.7  | 58.4  | 59.3  | 55.4  | 50.8  | 46.9  | 44.5  |
|                                                                        | 10%         | DB     | 45.1  | 44.3  | 45.8  | 50.5  | 57.1  | 60.8  | 62.9  | 63.9  | 59.6  | 52.8  | 47.6  | 45.0  |
|                                                                        |             | MCWB   | 43.1  | 41.2  | 41.7  | 44.3  | 50.0  | 54.0  | 57.3  | 58.2  | 55.2  | 49.8  | 45.2  | 42.9  |
|                                                                        | 0.4%        | WB     | 49.1  | 47.6  | 48.7  | 52.8  | 56.4  | 62.0  | 62.9  | 63.8  | 60.8  | 55.3  | 51.2  | 49.2  |
|                                                                        |             | MCDB   | 51.0  | 49.7  | 54.9  | 62.7  | 64.6  | 74.4  | 70.4  | 73.0  | 66.3  | 56.8  | 52.1  | 52.1  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 46.4  | 44.4  | 46.1  | 48.6  | 54.1  | 58.7  | 60.7  | 61.6  | 58.5  | 53.4  | 49.3  | 46.5  |
|                                                                        |             | MCDB   | 47.7  | 46.9  | 50.1  | 55.1  | 61.4  | 66.1  | 66.4  | 67.6  | 62.2  | 54.8  | 51.0  | 48.1  |
|                                                                        | 5%          | WB     | 44.8  | 43.0  | 44.1  | 46.8  | 52.4  | 56.6  | 59.4  | 60.2  | 57.2  | 51.7  | 47.2  | 44.9  |
|                                                                        |             | MCDB   | 46.2  | 45.5  | 47.2  | 52.2  | 58.7  | 62.3  | 63.8  | 65.1  | 60.2  | 53.2  | 49.2  | 46.5  |
|                                                                        | 10%         | WB     | 43.5  | 41.4  | 42.4  | 45.5  | 51.0  | 55.0  | 58.1  | 59.1  | 55.9  | 50.3  | 45.7  | 42.8  |
|                                                                        |             | MCDB   | 45.2  | 44.1  | 45.4  | 49.8  | 56.4  | 59.4  | 61.9  | 63.2  | 58.7  | 52.2  | 47.7  | 44.8  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 7.5   | 10.0  | 11.6  | 13.9  | 13.6  | 12.4  | 10.8  | 11.7  | 11.5  | 10.3  | 8.1   | 7.6   |
|                                                                        |             | MCDBR  | 8.2   | 11.3  | 15.7  | 20.9  | 21.4  | 19.8  | 16.2  | 19.0  | 17.3  | 12.5  | 8.1   | 8.8   |
|                                                                        |             | MCWBR  | 6.6   | 8.0   | 10.0  | 12.2  | 11.8  | 10.7  | 9.0   | 10.7  | 10.1  | 8.7   | 6.8   | 7.0   |
|                                                                        | 5% WB       | MDBR   | 7.7   | 9.6   | 13.2  | 17.1  | 17.4  | 16.6  | 13.7  | 15.2  | 12.3  | 8.8   | 7.7   | 8.4   |
|                                                                        |             | MCDBR  | 6.7   | 7.4   | 9.2   | 10.8  | 10.4  | 9.4   | 8.2   | 9.3   | 8.3   | 7.1   | 7.0   | 7.2   |
|                                                                        |             | MCWBR  | 6.7   | 7.4   | 9.2   | 10.8  | 10.4  | 9.4   | 8.2   | 9.3   | 8.3   | 7.1   | 7.0   | 7.2   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.305 | 0.313 | 0.344 | 0.362 | 0.366 | 0.367 | 0.379 | 0.365 | 0.366 | 0.340 | 0.315 | 0.301 |
|                                                                        | taud        |        | 2.340 | 2.473 | 2.386 | 2.373 | 2.416 | 2.448 | 2.452 | 2.517 | 2.460 | 2.435 | 2.306 | 2.242 |
|                                                                        | Ebn at Noon |        | 196   | 246   | 261   | 273   | 278   | 279   | 272   | 270   | 252   | 229   | 186   | 162   |
|                                                                        | Edh at Noon |        | 16    | 21    | 29    | 33    | 34    | 33    | 33    | 29    | 27    | 22    | 17    | 14    |
|                                                                        | RadAvg      |        | 145   | 387   | 706   | 1159  | 1463  | 1522  | 1354  | 1175  | 765   | 416   | 190   | 112   |
| (45) All-Sky Solar Radiation                                           | RadStd      |        | 20    | 73    | 101   | 79    | 157   | 136   | 178   | 122   | 107   | 57    | 26    | 20    |

## Historical Trends

|                         |  | Heating |        | Cooling |       |       | Degree-Days |       |       |       |     |
|-------------------------|--|---------|--------|---------|-------|-------|-------------|-------|-------|-------|-----|
|                         |  | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |     |
| Station Only            |  | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m) |
|                         |  | N/A     | N/A    | N/A     | N/A   | +1.04 | +1.08       | N/A   | -299  | +155  | N/A |
| Regional (14 neighbors) |  | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | +115  | N/A |

Nomenclature: See separate page